

Testing Report: Custom System Calls

Advances in OS Design — Autumn 2026–27, Assignment 2

Group Number: 04

Environment: Ubuntu 24.04.4 LTS, Linux Kernel 6.1.6 (compiled from source), GCC, x86_64

Task A: `proc_info`

#	Test Case	Input	Expected	Result
1	PID mode (self)	<code>get_proc_info(getpid())</code>	Valid info returned	PASS
2	PID mode (init)	<code>get_proc_info(1)</code>	PPID=0, state=1	PASS
3	System-wide mode	<code>get_system_info()</code>	Consistent counts	PASS
4	Invalid PID	<code>syscall(451, 99999, 1, ...)</code>	ESRCH	PASS
5	Invalid flags	<code>syscall(451, 1, 99, ...)</code>	EINVAL	PASS
6	Buffer too small	<code>syscall(451, 1, 1, buf, 1)</code>	ENOSPC	PASS

Key verification (Test 3): RT (57) + Fair (270) = 327 = Total Processes, confirming consistent counting. Only 1 process in CFS runqueue (the test program itself), as expected on an idle system.

Task B: `register_pid`, `fetch`, `deregister`

Two child processes were spawned via `fork()` to generate context switches through busy work and `usleep()` calls. Processes were deregistered one by one to verify cumulative counts decrease.

#	Test Case	Expected	Vol / Invol CS	Result
1	Register self (PID 3112)	Returns 0	—	PASS
2	Register PID 1	Returns 0	—	PASS
3	Register children (3113, 3114)	Returns 0	—	PASS
4	Fetch (4 monitored)	Cumulative counts	9026 / 1028	PASS
5	Deregister child 1	Returns 0	—	PASS
6	Fetch (3 monitored)	Lower counts	9015 / 1026	PASS
7	Deregister child 2	Returns 0	—	PASS
8	Fetch (2 monitored)	Lower counts	9004 / 1022	PASS
9	Deregister self	Returns 0	—	PASS
10	Fetch (PID 1 only)	Lowest counts	9002 / 1022	PASS
11	Deregister PID 1	Returns 0	—	PASS
12	Invalid PID (−1)	EINVAL	—	PASS
13	Deregister non-existent	ESRCH	—	PASS
14	Double deregister	ESRCH	—	PASS
15	Duplicate register	EEXIST	—	PASS

Key verification: Context switch counts show a clear downward trend as processes are deregistered (9026 → 9015 → 9004 → 9002), confirming that registration and deregistration correctly modify the monitored list. Thread-level counting is validated by the non-trivial contribution of child processes. Concurrency safety is ensured via mutex locking on all list operations.

All 21 tests passed. Both system calls behave as specified in the assignment.